IoT · EMBEDDED SYSTEMS PROJECT

Arduino-Based Smart Car Parking System

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01

Introduction

Automating the way we park smarter, faster, and stress-free.

With the increasing number of vehicles, finding a parking space has become a major challenge in cities, shopping malls, hospitals, and educational institutions. Traditional parking systems often cause traffic congestion, wasted time, and inefficient use of available spaces.

An Arduino-based Smart Car Parking System is an IoT project that automates parking by detecting available slots and controlling vehicle entry and exit. It displays the number of free spaces and automatically opens or closes the gate using a servo motor making parking more efficient and user-friendly.

02

Problem Statement

Conventional parking systems require manual monitoring and do not provide real-time information about available parking spaces. As a result, drivers spend unnecessary time searching for parking leading to traffic congestion, fuel wastage, and frustration.

The Need

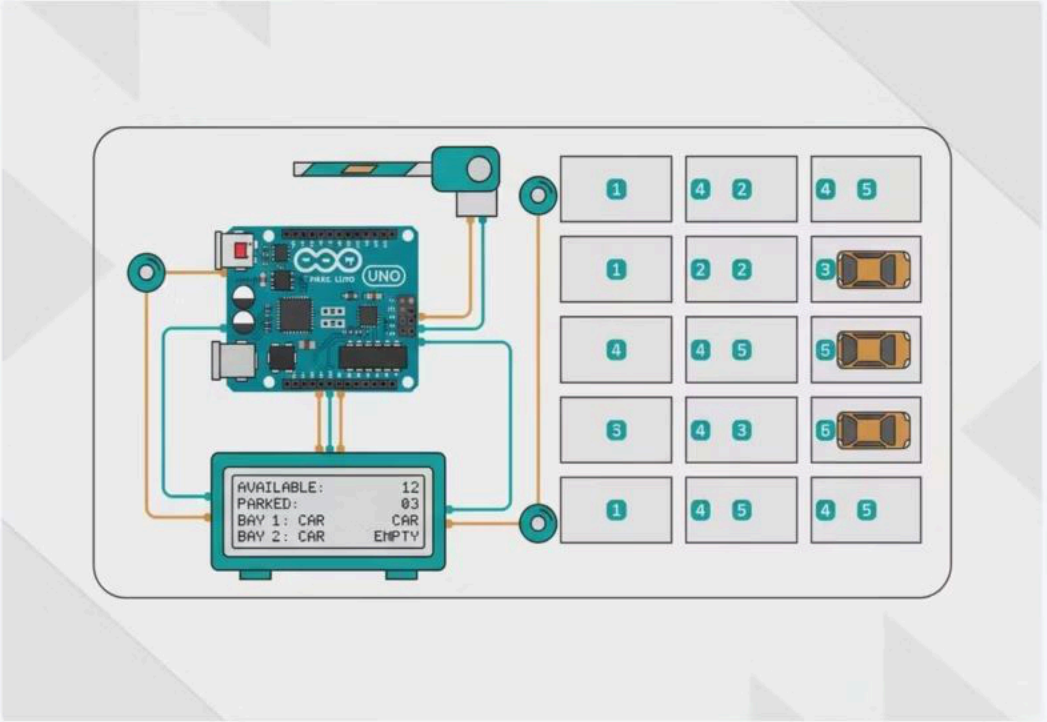
A low-cost, automated parking management system that monitors slot availability and regulates vehicle entry efficiently.

Explanation of the Idea

The system uses an Arduino Uno as the main controller. IR sensors at the entrance and at individual slots detect vehicles. When a car arrives, the system checks whether any space is available.

OPERATION WORKFLOW

- If a slot is available, the servo motor opens the gate for the vehicle.
- If all slots are occupied, the gate stays closed and the LCD shows "Parking Full".
- Sensors update the number of available spaces in real time.
- The LCD continuously displays current parking status.



This project demonstrates embedded systems, sensors, automation, and IoT concepts to improve parking management while reducing human effort and traffic congestion.

04

Technologies Used

- Arduino UNO R3 – 1
- 20×4 LCD Display – 1
- I2C LCD Module – 1
- IR Obstacle Sensor Modules – 8
- SG90 9g Mini Servo Motor – 1
- Male Header Pins – 1 set
- Female Header Pins – 1 set
- Normal Connecting Wires – 1 set
- Female DC Power Jack – 1
- 5V 2A Power Adapter – 1
- USB Cable for Arduino – 1
- Cardboard/Foam Board for Model – 1 set

Methodology

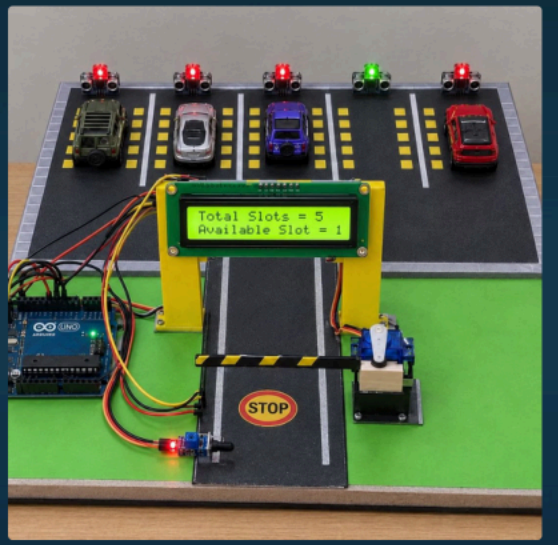
- Study parking problems and define system requirements
- Select Arduino Uno and the IR, servo, and LCD modules
- Design the circuit and wire the sensors and actuators
- Write and test the control code in Arduino IDE (C/C++)
- Build the model and calibrate the sensors

Working

- IR sensors at the entrance and each slot detect arriving vehicles
- The Arduino checks whether any slot is available
- If a slot is free, the servo motor opens the gate
- If all slots are occupied, the gate stays closed and the LCD shows "Parking Full"
- The LCD continuously displays the number of available spaces

Results

- Real-time parking status is shown on the LCD
- The gate opens automatically only when a slot is free
- Less manual effort and reduced congestion at the entrance
- A working prototype demonstrates embedded systems and IoT concepts



Applications

Shopping Malls
Manages vehicle parking efficiently and reduces waiting time.

Office Buildings
Monitors parking availability for employees and guests.

Hospitals
Provides organized parking for patients, visitors, and staff.

Airports & Railway Stations
Assists in handling large volumes of vehicles.

Educational Institutions
Controls parking for students, faculty, and visitors.

Residential & Gated Communities
Automates entry and exit while monitoring occupancy.

Thank You